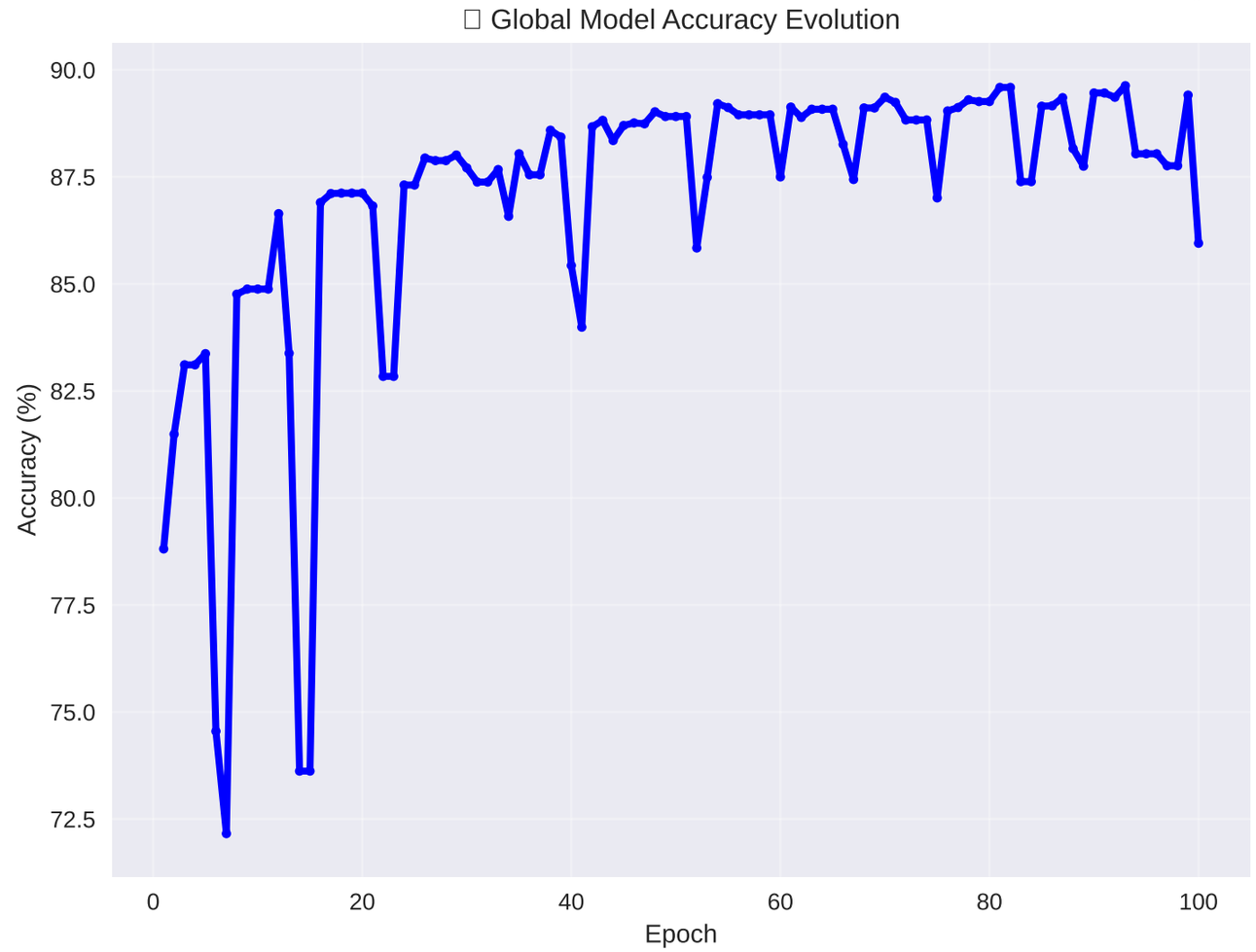
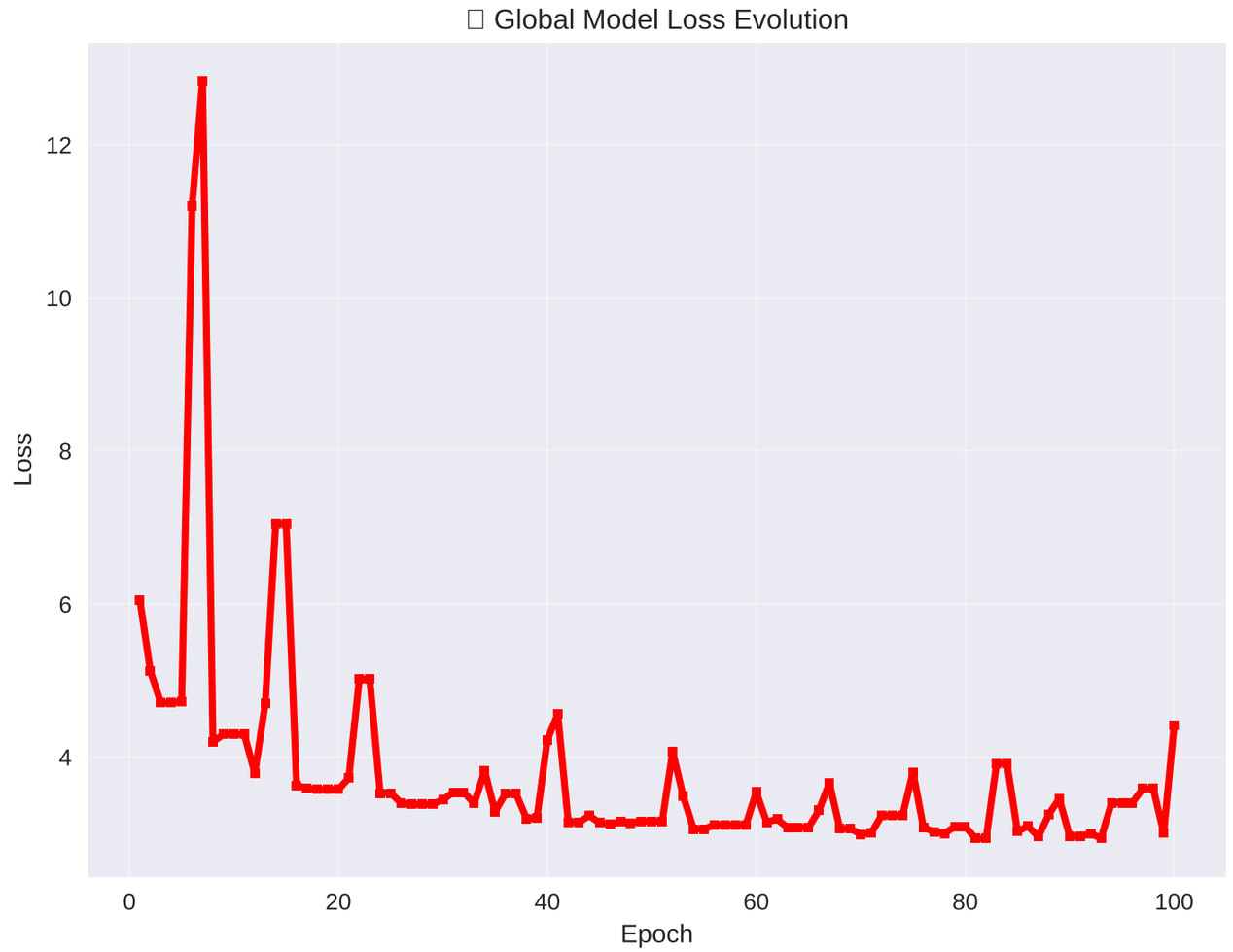
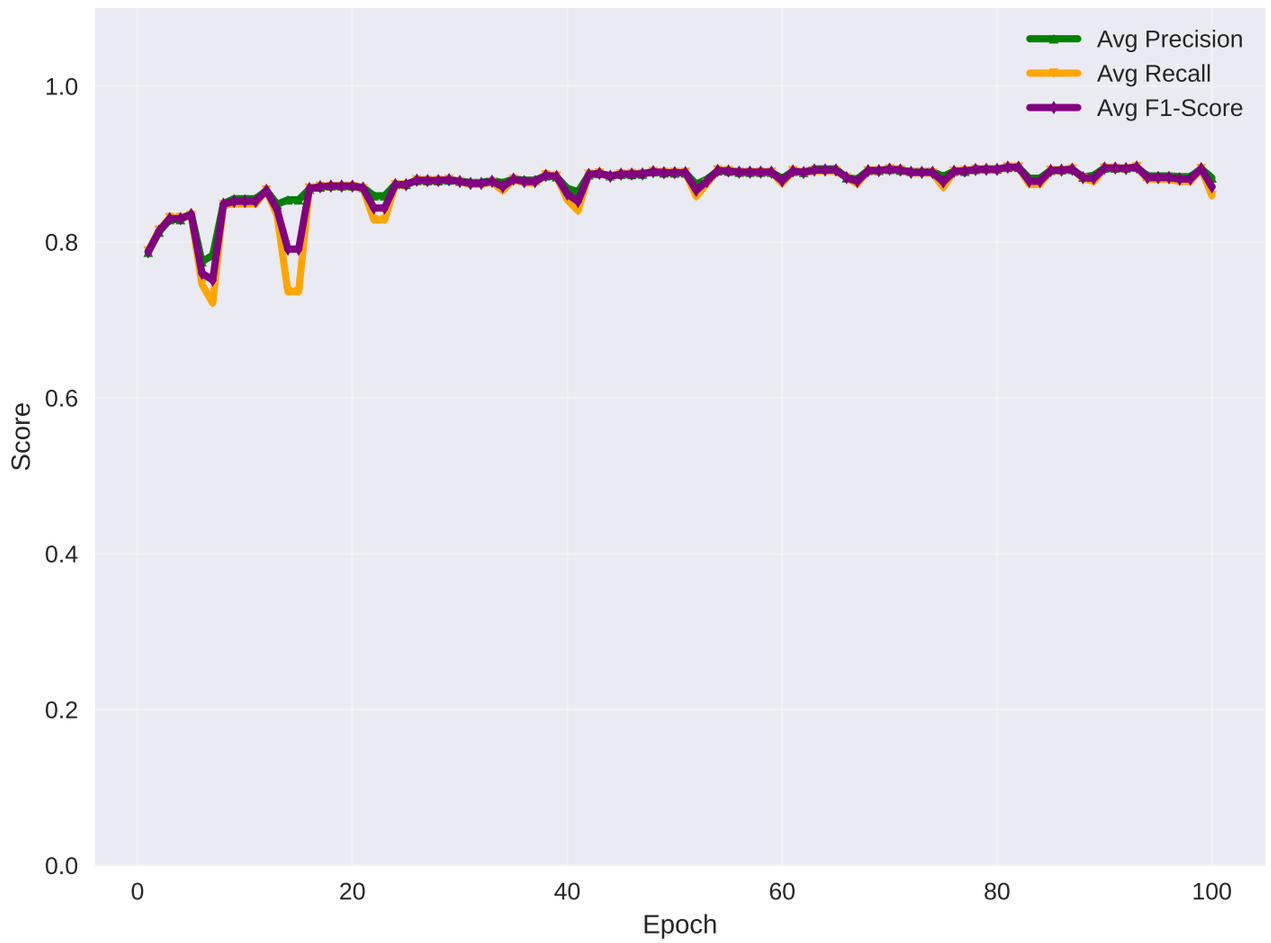
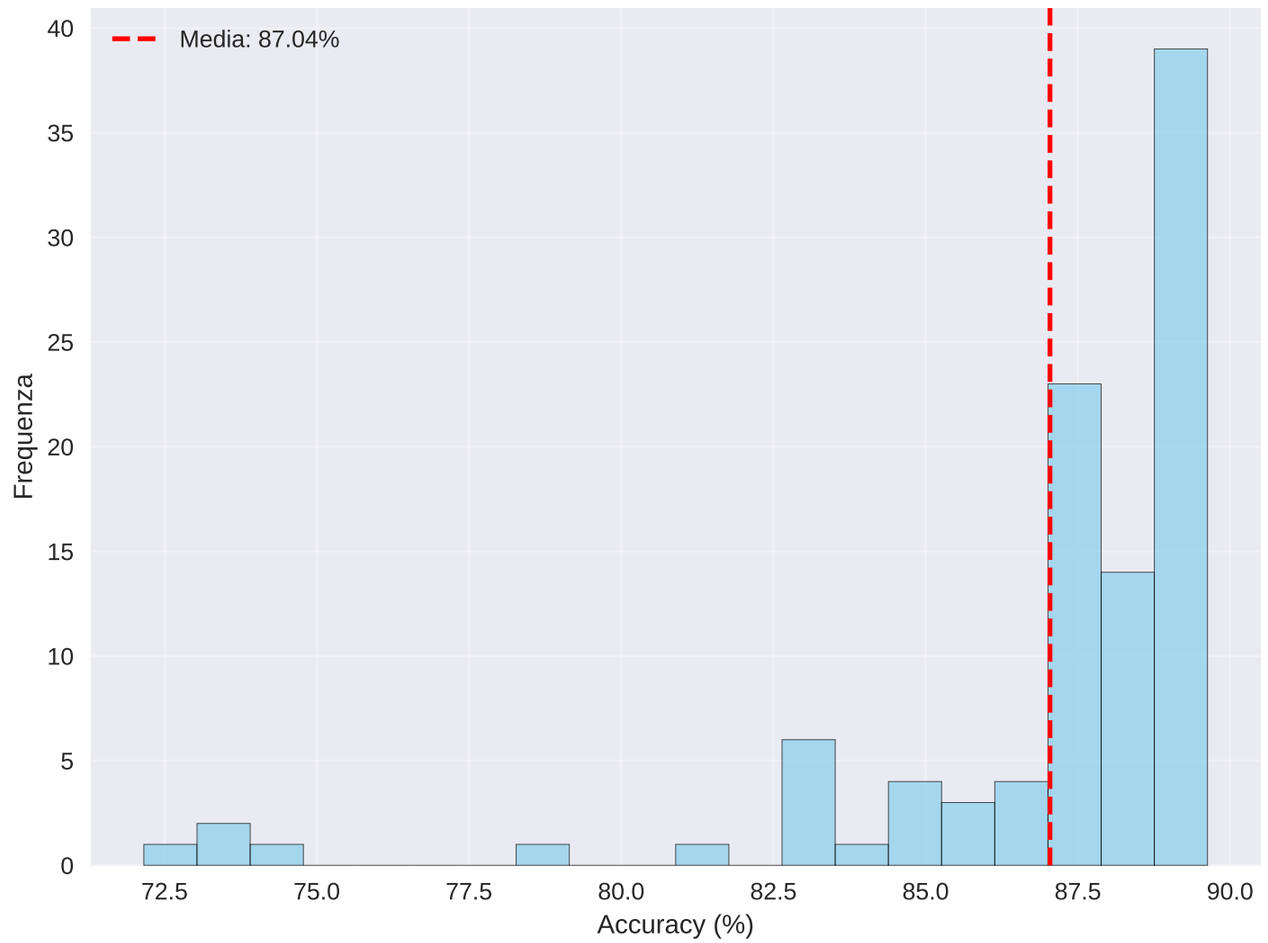
Federated Learning - Main Metrics Overview

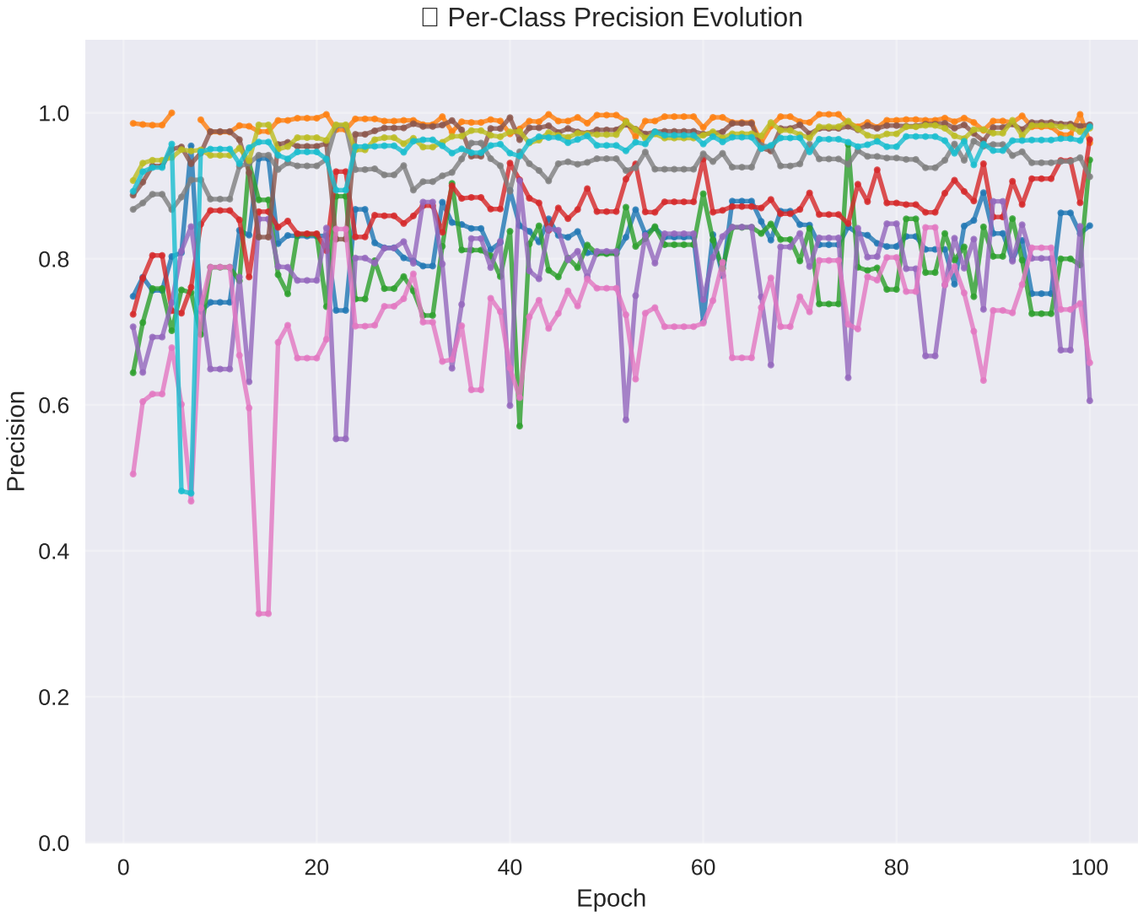


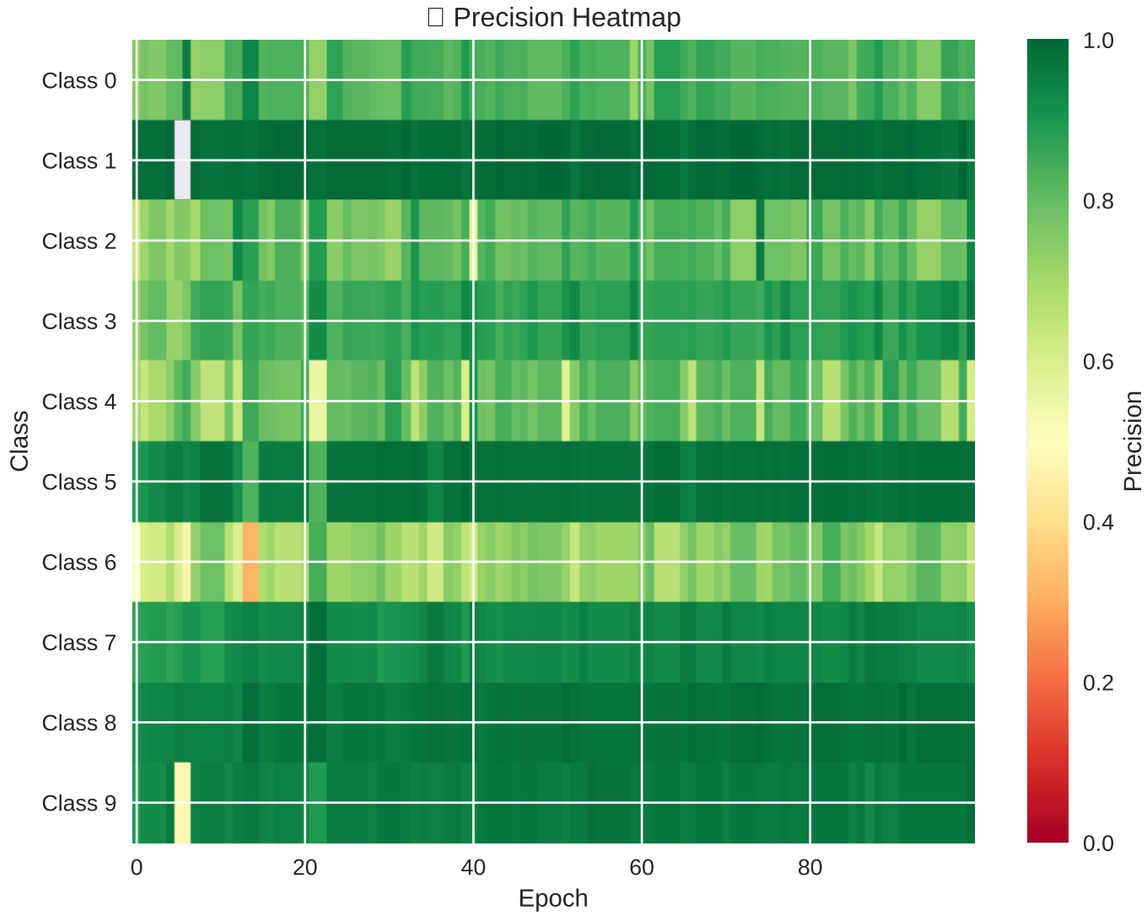


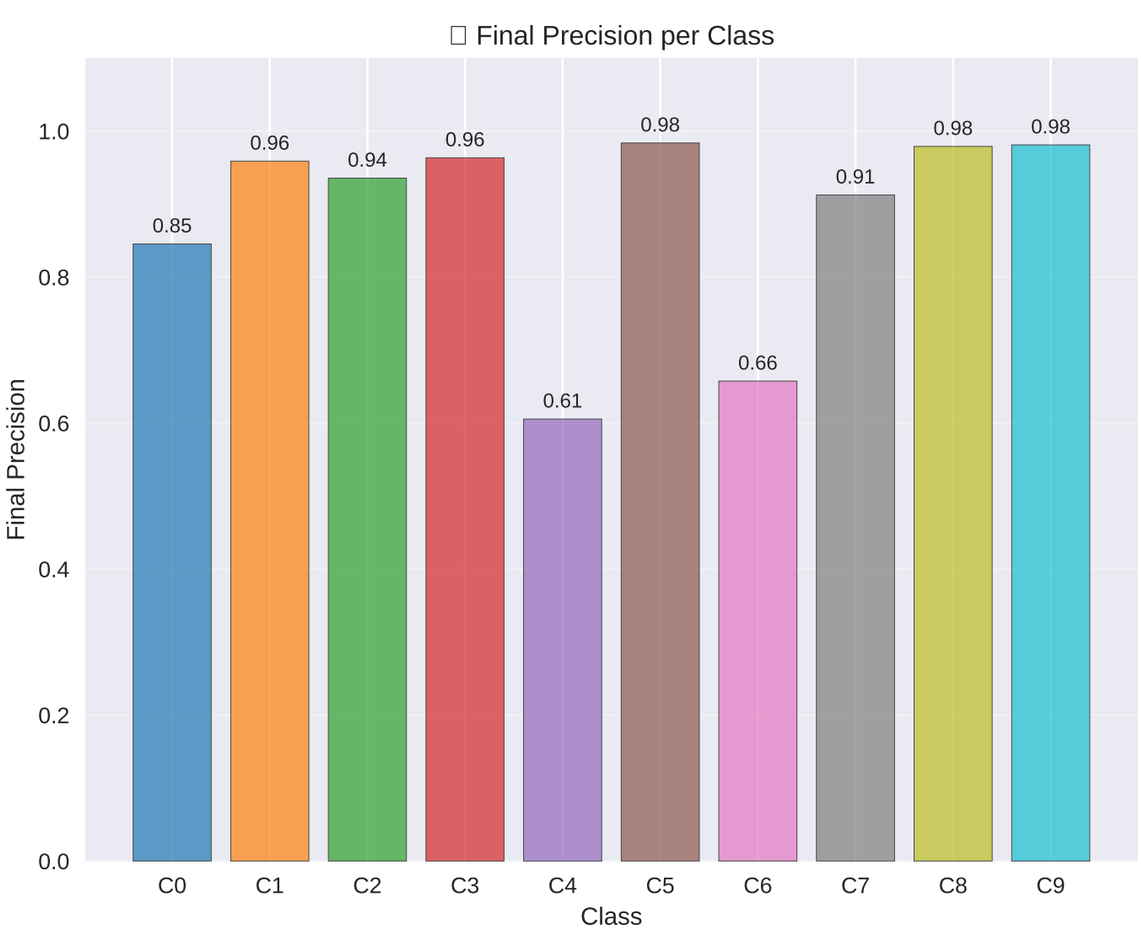


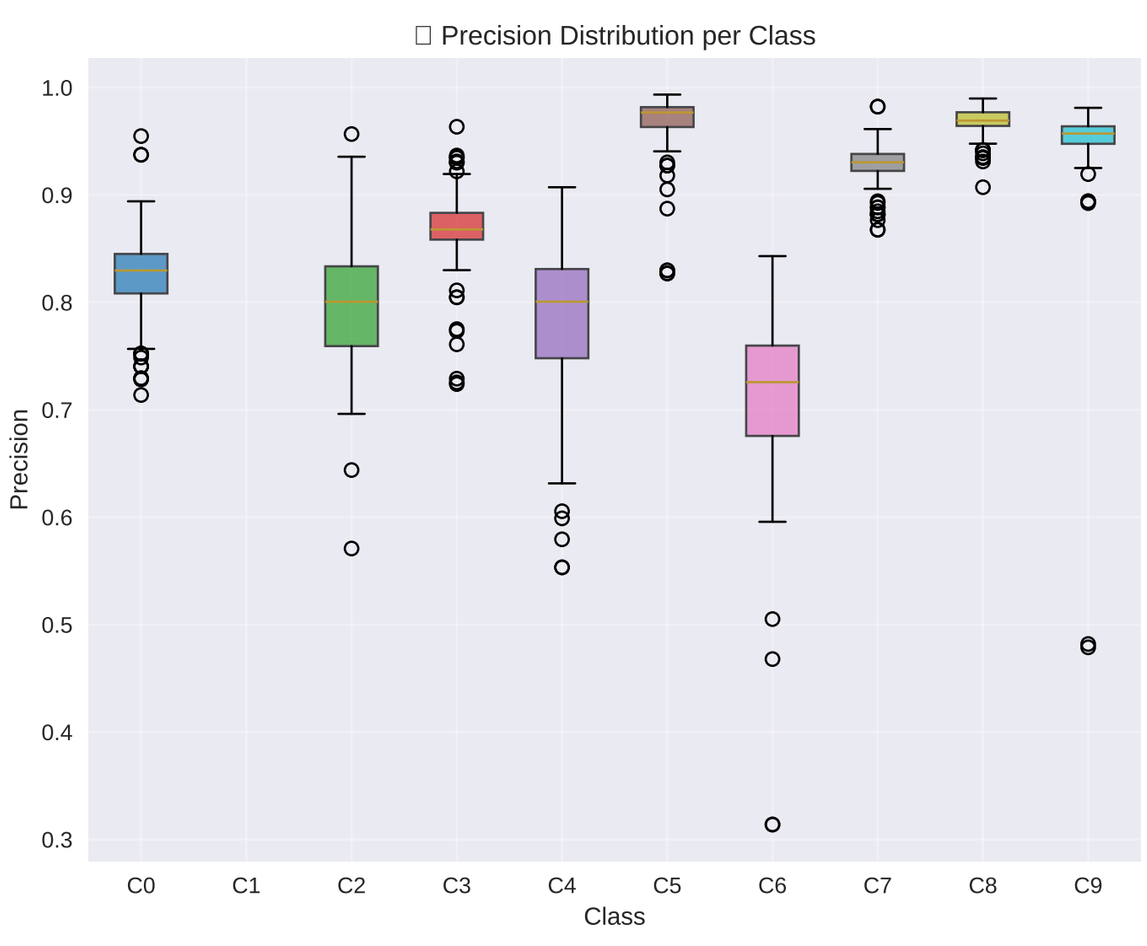


Per-Class Precision Analysis



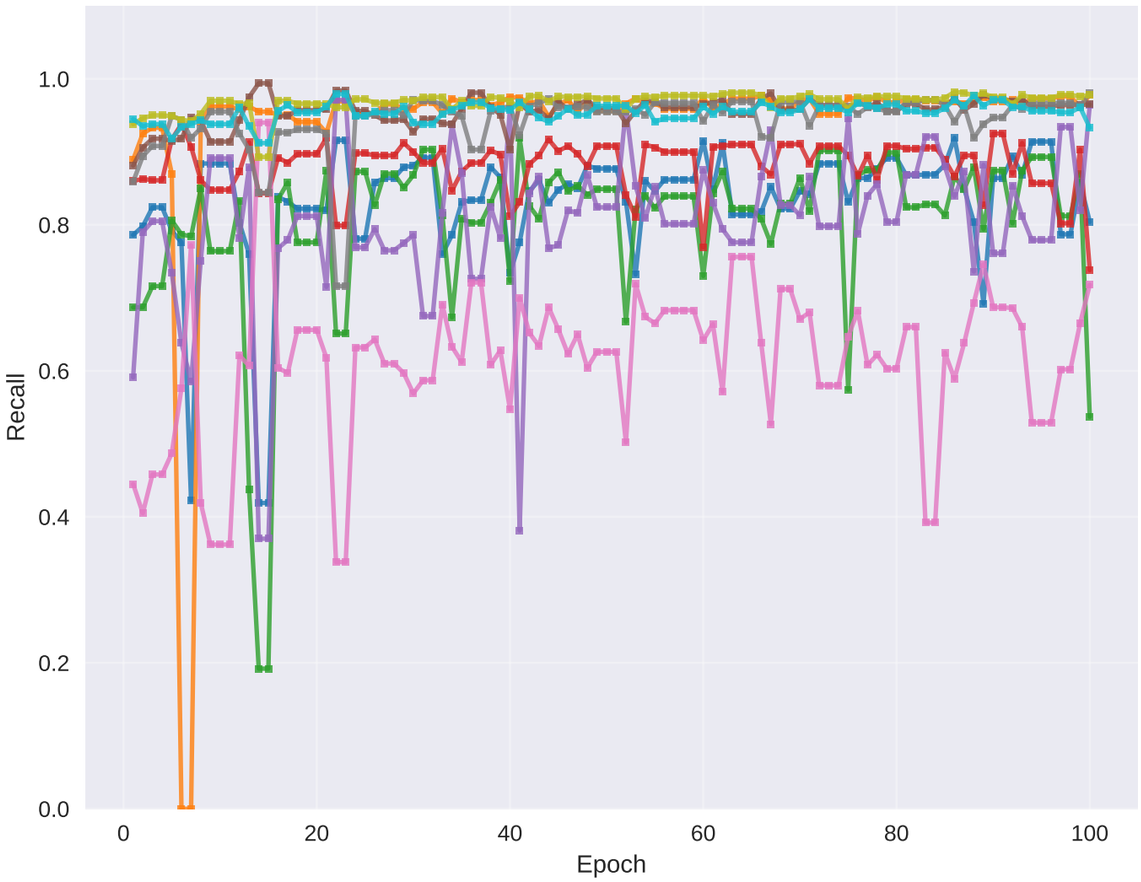




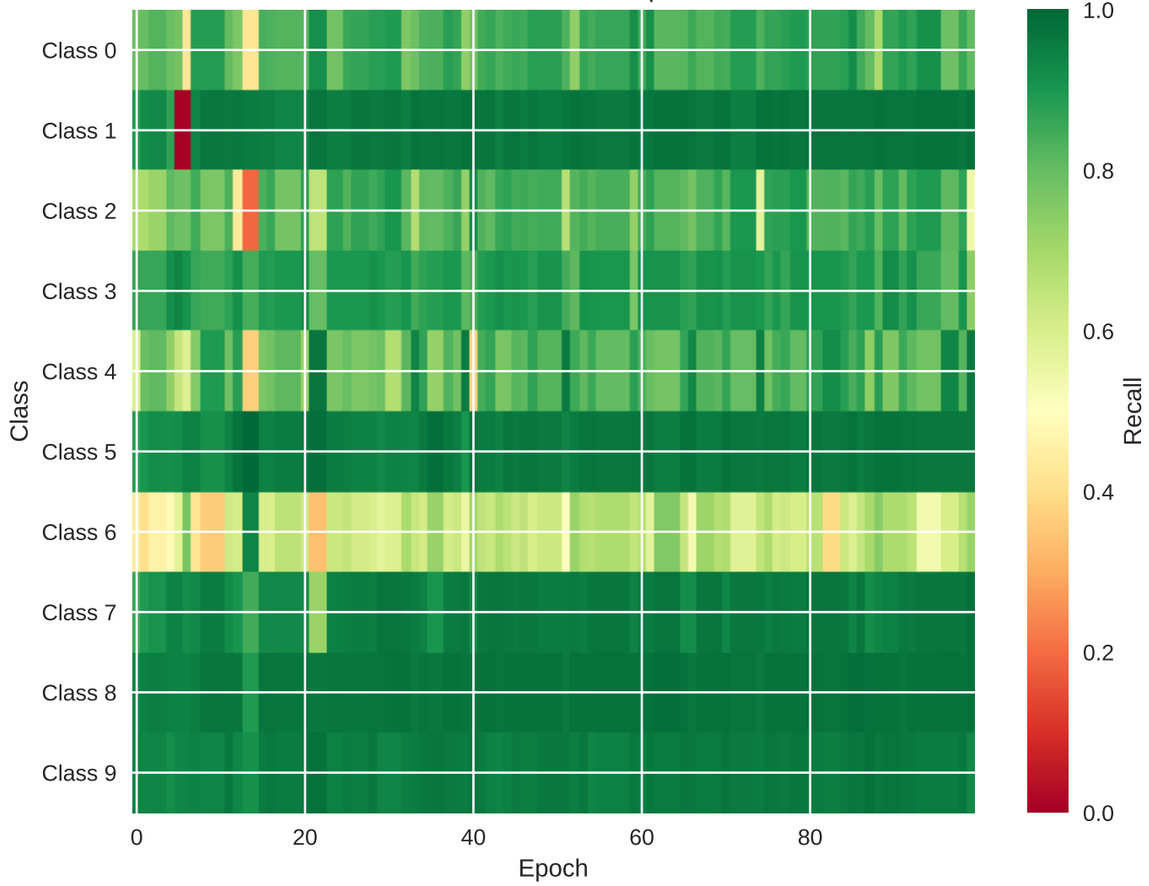


Per-Class Recall Analysis

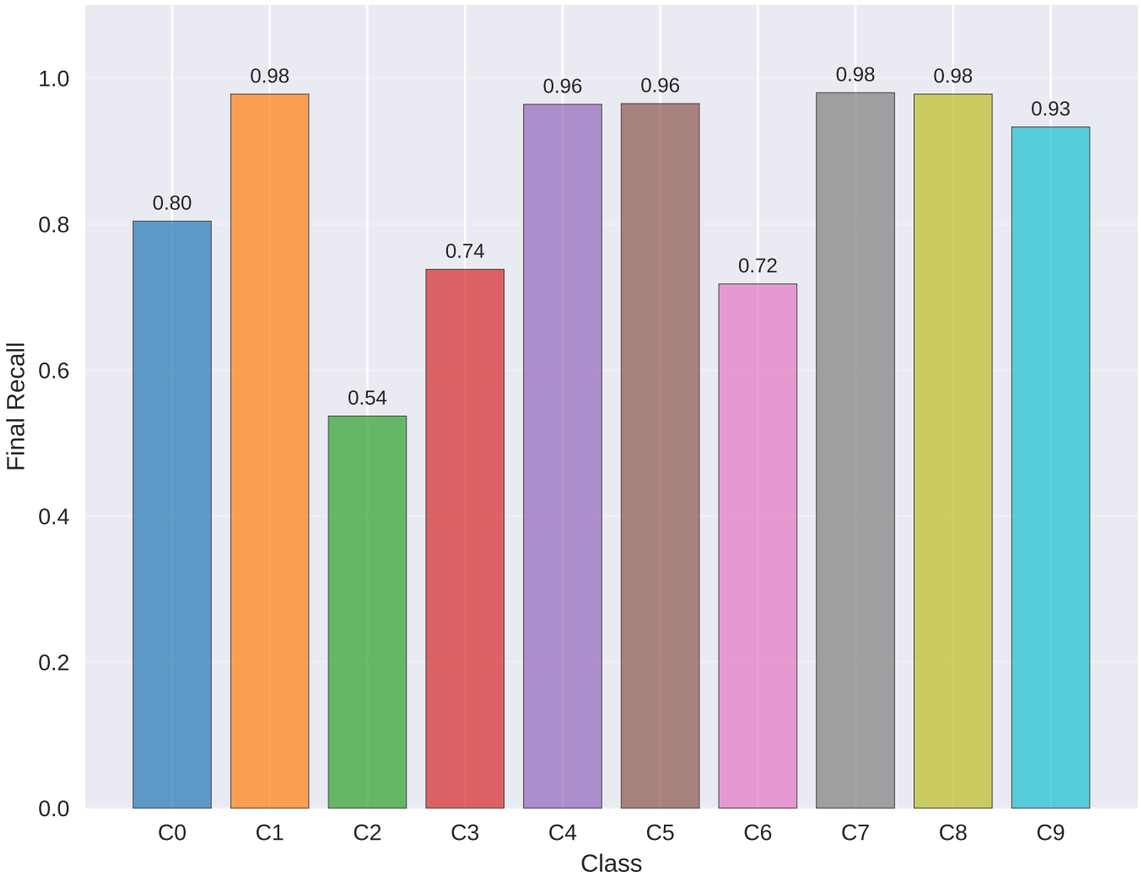
□ Per-Class Recall Evolution



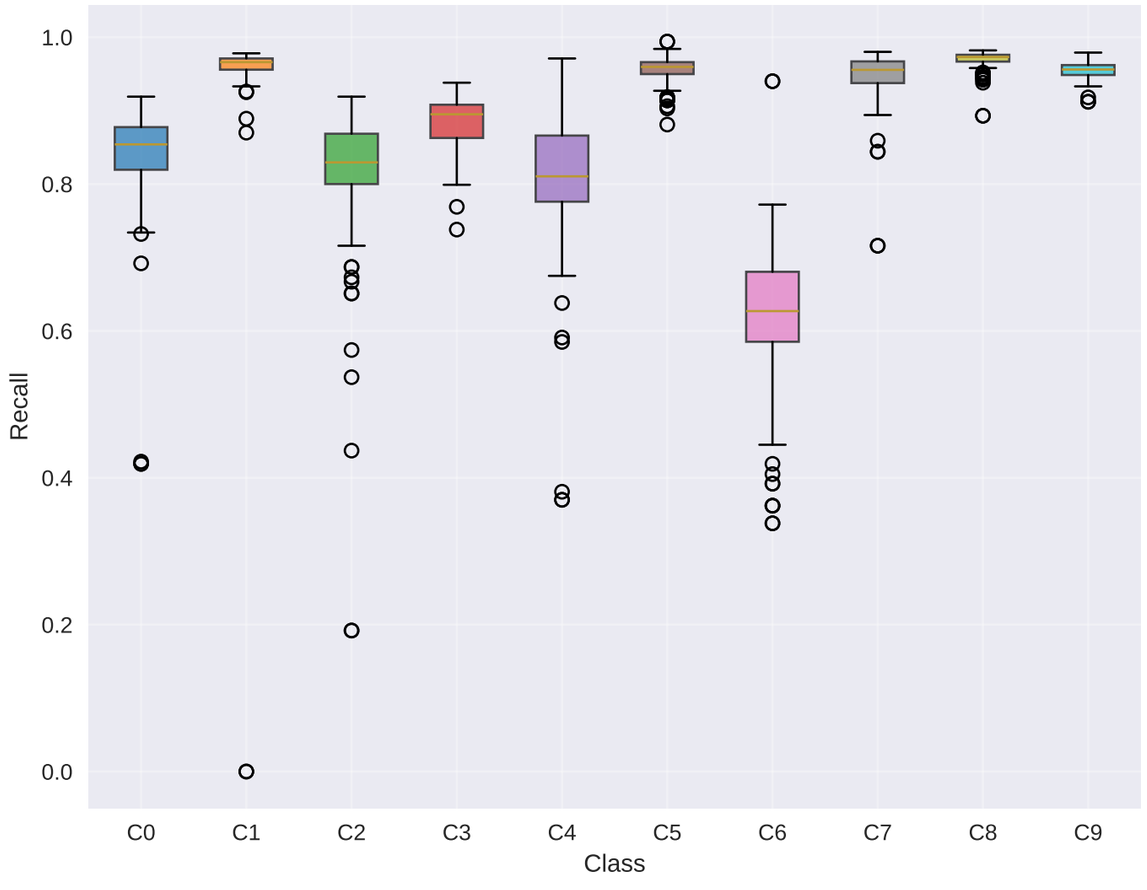
□ Recall Heatmap



□ Final Recall per Class

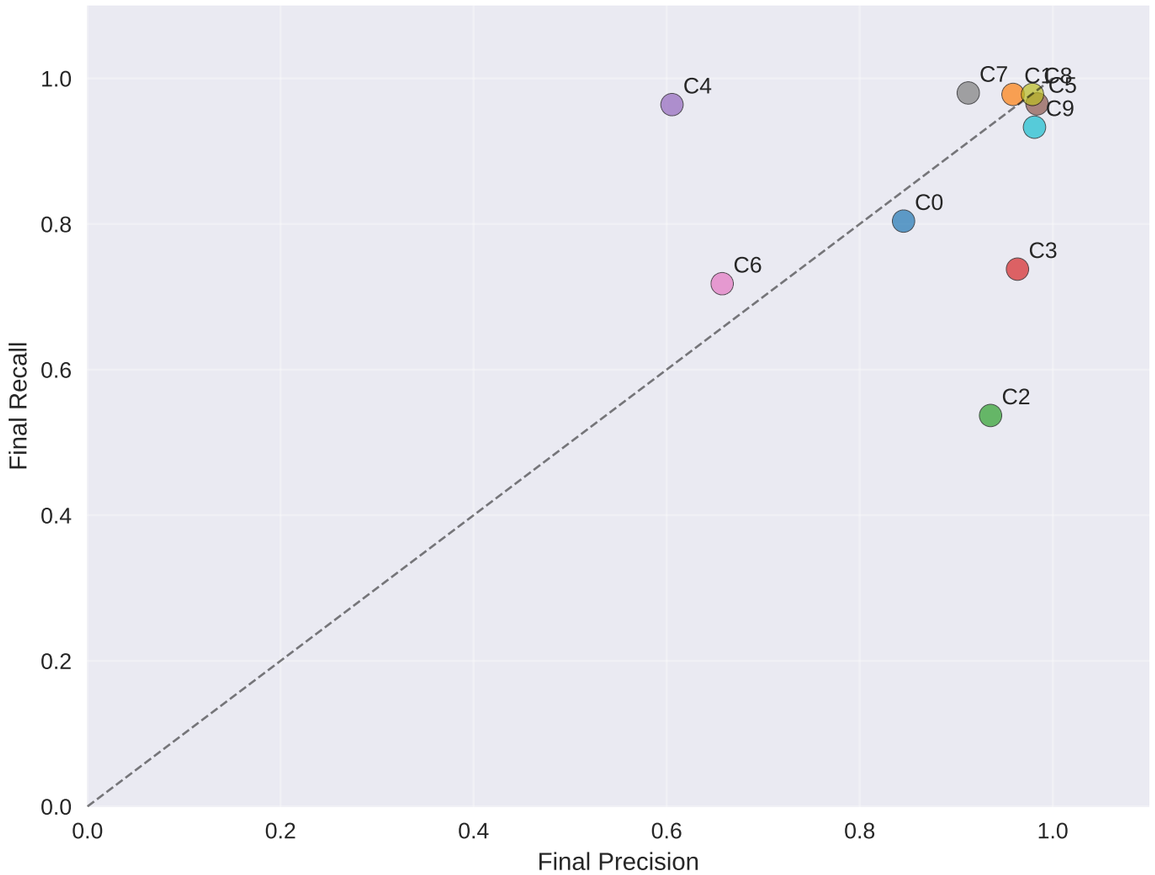


□ Recall Distribution per Class

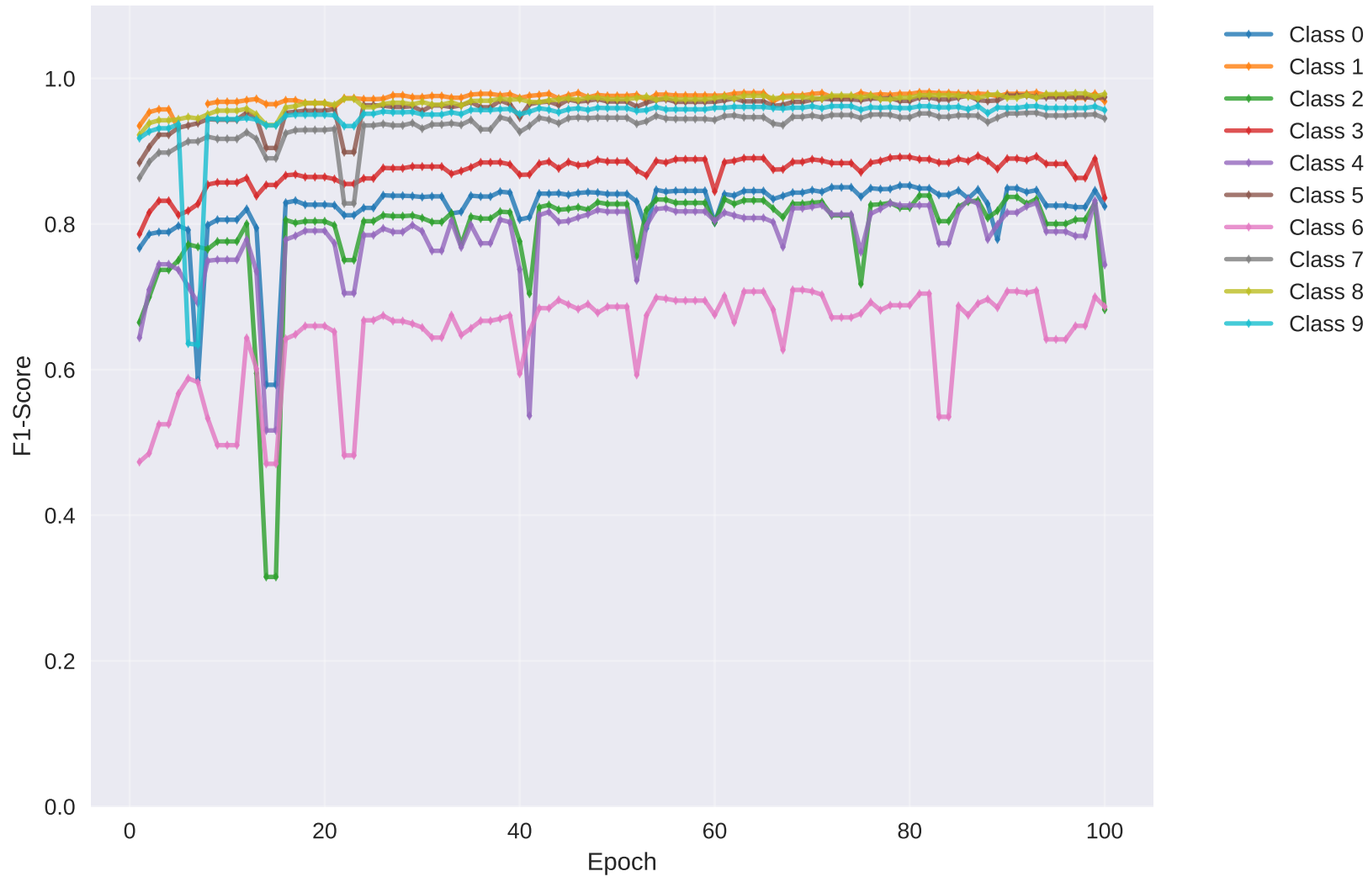


Comparative Analysis

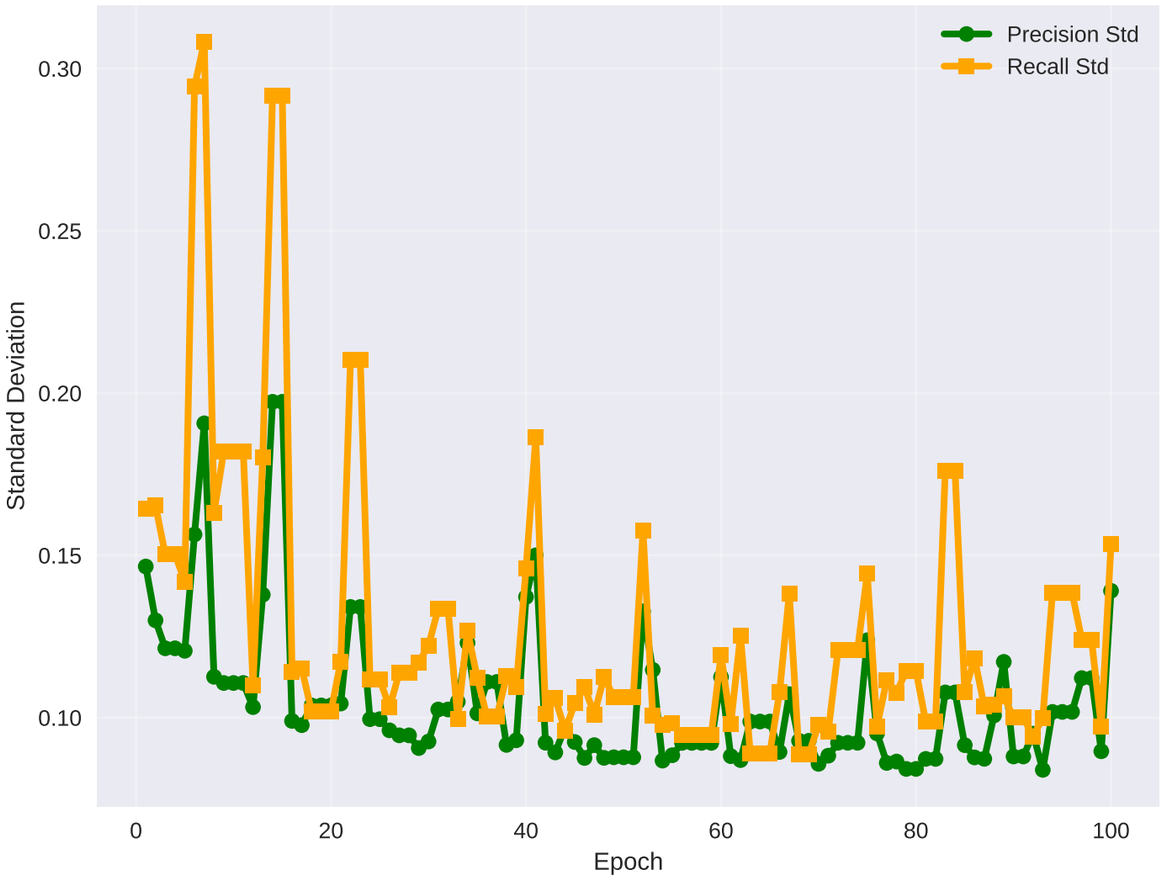
□ Precision vs Recall (Final Values)



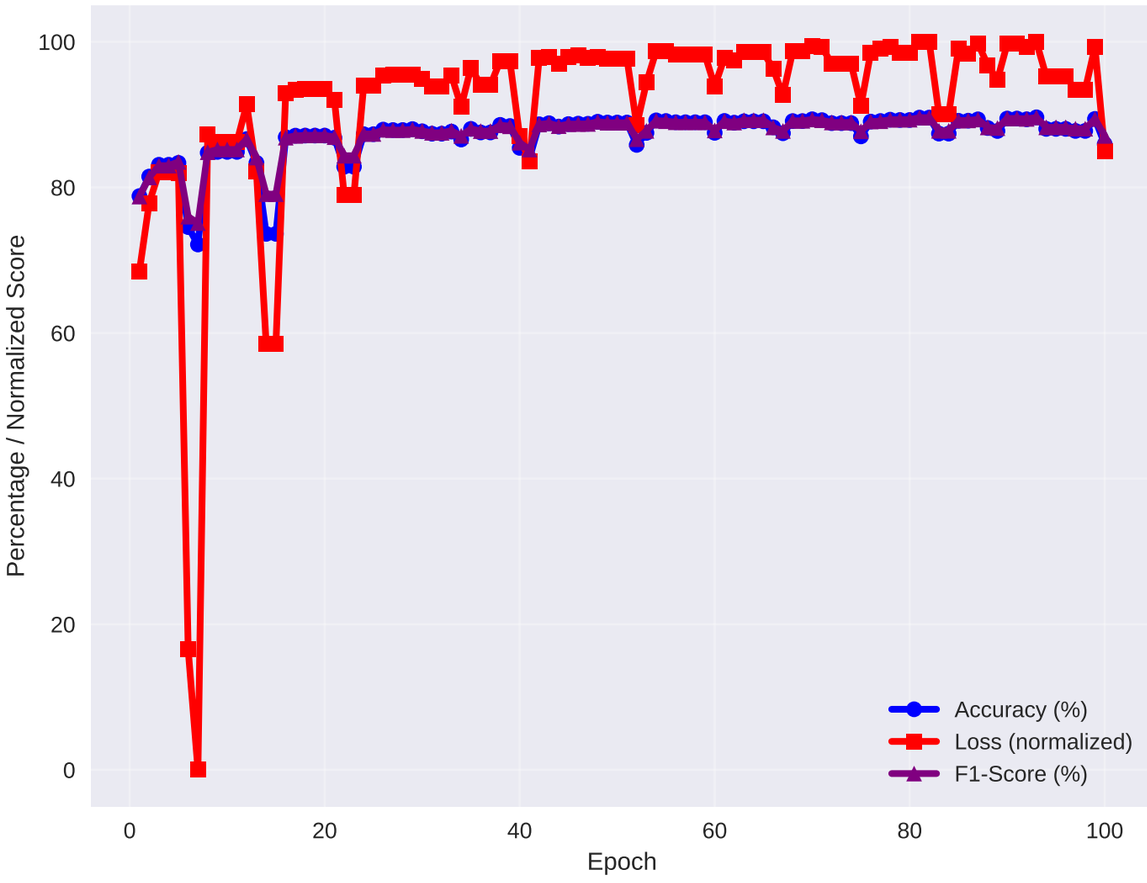
□ Per-Class F1-Score Evolution



□ Inter-Class Variance Evolution



□ Learning Curves Comparison



Performance Summary Report

```
FEDERATED LEARNING - PERFORMANCE SUMMARY
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FILE ANALIZZATO: 3000_results.csv
DATA ANALISI: 2025-09-10 13:24:21
EPOCHE TOTALI: 100

METRICHE FINALI:
• Accuracy: 85.95%
• Loss: 4.4211
• Average Precision: 0.882
• Average Recall: 0.859
• Average F1-Score: 0.871

EVOLUZIONE TRAINING:
• Miglioramento Accuracy: +7.14%
• Riduzione Loss: +1.6337
• Stabilità Accuracy (CV): 0.0394

PERFORMANCE PER CLASSE:
• Migliore Precision: Classe 5 (0.984)
• Peggiore Precision: Classe 4 (0.606)
• Migliore Recall: Classe 7 (0.980)
• Peggiore Recall: Classe 2 (0.537)

ANALISI DETTAGLIATA PER CLASSE:
• Classe 0: P=0.845, R=0.804, F1=0.824
• Classe 1: P=0.959, R=0.978, F1=0.968
• Classe 2: P=0.936, R=0.537, F1=0.682
• Classe 3: P=0.963, R=0.738, F1=0.836
• Classe 4: P=0.606, R=0.964, F1=0.744
• Classe 5: P=0.984, R=0.965, F1=0.974
• Classe 6: P=0.658, R=0.718, F1=0.686
• Classe 7: P=0.912, R=0.980, F1=0.945
• Classe 8: P=0.979, R=0.978, F1=0.978
• Classe 9: P=0.981, R=0.933, F1=0.956

VALUTAZIONE QUALITATIVA:
• Performance Globale: MEDIA
• Stabilità Training: MODERATAMENTE STABILE
• Balance Classes: SBILANCIATO

RACCOMANDAZIONI:
• Considerare aumento epoche o tuning iperparametri
• Implementare class balancing o weighted loss
```